

MA201: Multivariable Calculus

3D Coordinate Systems & Vectors — Lecture 01

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1. Three-Dimensional Coordinate Systems

To locate a point in space, we choose a fixed point O (the origin) and three mutually perpendicular directed lines through O called the coordinate axes: x -axis, y -axis, and z -axis.

Distance Formula in 3D Space

The distance $|P_1P_2|$ between points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is given by:

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Equation of a Sphere

An equation of a sphere with center $C(h, k, l)$ and radius r is:

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

Right-Hand Rule

Curl the fingers of your right hand in the direction of a 90° rotation from the positive x -axis to the positive y -axis; your thumb points along the positive z -axis.

2. Vectors in Three Dimensions

A three-dimensional vector is an ordered triple:

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$$

Its magnitude is:

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

Stewart 12.1 Example

Show that $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$ is the equation of a sphere, and find its center and radius.

Solution: Complete the square for x , y , and z :

$$(x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) = -6 + 4 + 9 + 1$$

$$(x + 2)^2 + (y - 3)^2 + (z + 1)^2 = 8$$

Sphere has center $(-2, 3, -1)$ and radius $r = \sqrt{8} = 2\sqrt{2}$.